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British Sign Language Detection Using Ultra-Wideband Radar Sensing and Residual Neural Network

[Umer Saeed](#), [Syed Aziz Shah](#), Yazeed Yasin Ghadi, Hira Hameed, Syed Ikram Shah, Jawad Ahmad, Qammer H. Abbas

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Abstract

This study represents a significant advancement in Sign Language Detection (SLD), a crucial tool for enhancing communication and fostering inclusivity among the hearing-impaired community. It innovatively combines radar

technology with deep learning techniques to develop a sophisticated, non-invasive SLD system. Traditional SLD methods often rely on cumbersome wearable devices or struggle with environmental inconsistencies. In contrast, this system utilizes the distinctive ability of radar to function effectively across various lighting conditions. The core of this research lies in its application to British Sign Language (BSL) detection, employing advanced neural network architectures for real-time interpretation. A key highlight is the impressive 92% accuracy rate achieved in BSL recognition, utilizing the Residual Neural Network (ResNet) model. This success is attributed to a comprehensive dataset and the strategic adaptation of ResNet for processing radar data. The fusion of radar technology with deep learning in this context not only marks a novel approach in the field but also establishes this research as a foundational contribution to the realm of SLD. Its implications extend beyond technical achievement, offering a more accessible and inclusive communication alternative for the hearing-impaired.

Original language English

Pages (from-to) (In-Press)

Number of pages 8

Journal [IEEE Sensors Journal](#)

Volume (In-Press)

DOIs • <https://doi.org/10.1109/jsen.2024.3364389>

Publication status E-pub ahead of print - 15 Feb 2024

Funder

Engineering and Physical Research Council (EPSRC) (Grant Number: EP/W037076/1)

Keywords

- British sign language
- radar sensing
- gesture-reading
- contactless detection
- ResNet

Access to Document

- [10.1109/jsen.2024.3364389](https://doi.org/10.1109/jsen.2024.3364389)

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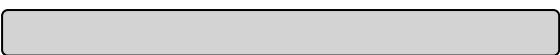
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[Saeed, U., Shah, S. A.](#), Ghadi, Y. Y., Hameed, H., Shah, S. I., Ahmad, J., & Abbas, Q. H. (2024). [British Sign Language Detection Using Ultra-Wideband Radar Sensing and Residual Neural Network](#). **IEEE Sensors Journal**, (In-Press), (In-Press). Advance online publication. <https://doi.org/10.1109/jsen.2024.3364389>

[British Sign Language Detection Using Ultra-Wideband Radar Sensing and Residual Neural Network](#). / [Saeed, Umer; Shah, Syed Aziz](#); Ghadi, Yazeed Yasin et al.

In: [IEEE Sensors Journal](#), Vol. (In-Press), 15.02.2024, p. (In-Press).

Research output: Contribution to journal › Article › peer-review

[Saeed, U., Shah, SA](#), Ghadi, YY, Hameed, H, Shah, SI, Ahmad, J & Abbas, QH 2024, 'British Sign Language Detection Using Ultra-Wideband Radar Sensing and Residual Neural Network', **IEEE Sensors Journal**, vol. (In-Press), pp. (In-Press). <https://doi.org/10.1109/jsen.2024.3364389>

[Saeed U, Shah SA](#), Ghadi YY, Hameed H, Shah SI, Ahmad J et al. [British Sign Language Detection Using Ultra-Wideband Radar Sensing and Residual Neural Network](#). [IEEE Sensors Journal](#). 2024 Feb 15;(In-Press):(In-Press). Epub 2024 Feb 15. doi: 10.1109/jsen.2024.3364389

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@article{dafd8031594c4b46bcfdf490befda71e,

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title = "British Sign Language Detection Using Ultra-Wideband Radar
Sensing and Residual Neural Network",
abstract = "This study represents a significant advancement in Sign
Language Detection (SLD), a crucial tool for enhancing communication and
fostering inclusivity among the hearing-impaired community. It innovatively
combines radar technology with deep learning techniques to develop a
sophisticated, non-invasive SLD system. Traditional SLD methods often rely
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alternative for the hearing-impaired.",
keywords = "British sign language, radar sensing, gesture-reading,
contactless detection, ResNet",
author = "Umer Saeed and Shah, {Syed Aziz} and Ghadi, {Yazeed Yasin}
and Hira Hameed and Shah, {Syed Ikram} and Jawad Ahmad and Abbas,
{Qammer H.}",
year = "2024",
month = feb,
day = "15",
doi = "10.1109/jsen.2024.3364389",
language = "English",
volume = "(In-Press)",
pages = "(In--Press)",
journal = "IEEE Sensors Journal",
issn = "1530-437X",
publisher = "Institute of Electrical and Electronics Engineers",
}

```

TY - JOUR

T1 - British Sign Language Detection Using Ultra-Wideband Radar Sensing
and Residual Neural Network

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AU - Shah, Syed Aziz

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PY - 2024/2/15

Y1 - 2024/2/15

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U2 - 10.1109/jsen.2024.3364389

DO - 10.1109/jsen.2024.3364389

M3 - Article

SN - 1530-437X

VL - (In-Press)

SP - (In-Press)

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